302

BTS – III(SS) – 06.13 – 0726

B.Tech. Degree III Semester Special Supplementary Examination June 2013

IT/ME/EC/EB/EI 302 ELECTRICAL TECHNOLOGY
(2006 Scheme)

Time : 3 Hours | Maximum Marks : 100

PART A
(Answer *ALL* questions)

($8 \times 5 = 40$)

I. (a) The no load ratio required in a single phase 50Hz transformer is 6600/600V. If the maximum value of flux in the core is to be about 0.08 wb. find the number of turns in each winding.

(b) Derive the condition for maximum efficiency in a transformer.

(c) What are the conditions for building up of voltage in a dc shunt generator?

(d) Series motor never started on no load. Explain why.

(e) Derive the speed frequency equation of an alternator.

(f) Draw the torque-slip characteristics of a 3-phase induction motor for different values of rotor resistance.

(g) Explain what do you mean by skin effect.

(h) Write short notes on load despatching.

PART B

($4 \times 15 = 60$)

II. (a) Draw the phasor diagram of a transformer on no load and also when the secondary is connected to an inductive load.

(b) A single phase 3KVA, 230/115V 50Hz transformer has the following constants.
Resistance: Primary = 0.3Ω Secondary = 0.09Ω
Reactance: Primary = 0.4 Ω Secondary = 0.1 Ω
What would be the readings of the instruments when the transformer is connected for SC test, if supply is given to HV side?

OR

III. (a) What do you mean by CT and PT? Where they are used?

(b) Prove that for the same capacity and voltage ratio, an auto transformer requires less copper than an ordinary transformer.

IV. (a) Derive the emf equation of a dc generator.

(b) A 4 pole dc shunt generator with a shunt field resistance of 100 ohms and an armature resistance of 1 ohm, has 378 wave connected conductors in its armature. The flux per pole is 0.02 wb. If a load resistance of 10 ohms is connected across the armature terminals and the generator is driven at 1000 rpm, calculate power absorbed by load.

OR

(P.T.O.)

V. (a) Draw and explain the characteristics of a dc series motor.

(b) A 200V dc shunt motor takes 4A at no load when running at 700 rpm. The field resistance is 100Ω. The resistance of armature at standstill gave a drop of 6V across armature terminals when 10A passed through it. Full load power input of the motor is 8KW. Calculate (i) speed on full load (ii) efficiency.

VI. A 3-phase star connected, 1000KVA, 11000V alternator has rated current of 52.5A. The ac resistance of the winding per phase is 0.45ohm. The test results are given below.

OC test : Field current = 12.5A
Voltage between lines = 422V

SC test : Field current = 12.5A
Line current = 52.5A

Determine the full load voltage regulation of the alternator at (i) 0.8 pf lagging and (ii) 0.8 pf leading.

OR

VII. (a) A 6 pole alternator running at 1000 rpm supplies an 8 pole induction motor. What is the actual speed of the motor if the slip is 2.5%?

(b) Why single phase induction motor is not self starting? Explain two starting methods.

VIII. (a) With necessary schematic diagram explain the working of a hydroelectric power plant.

(b) Explain what do you mean by corona.

OR

IX. (a) Compare over head and underground distribution system.

(b) Draw the typical layout of a nuclear power station.

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